# Marco Schouten

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## EDUCATION

**KTH Royal Institute of Technology** — Stockholm, Sweden
Master of Science in Machine Learning, Grade 4.84/5 — *Aug 2020 - Sep 2022*

**Delft University of Technology** — Delft, Netherlands
Exchange (Master) in Computer Science — *Aug 2021 - Jan 2022*

**University of Genoa** — Genoa, Italy
Bachelor of Science in Computer Science Engineering, Grade 109/110 — *Sep 2017- Jul 2020*

## EXPERIENCE

**Aptiv PLC** — Wuppertal, Germany
*Machine Learning Researcher* — *Mar 2023 - Present*

• Collaborating with a team of engineers to develop a customized neural network architecture that improves centerline accuracy for high-definition (HD) maps.

**SMaRC - Swedish Maritime Robotic Center** — Stockholm, Sweden
*Master Thesis: Underwater vehicle pose estimation* — *Feb 2022 - Sep 2022*

• Developed and implemented a novel algorithm to improve the accuracy of sonar-based pose estimation for autonomous underwater vehicles, reducing drift and achieving accuracy up to 1m.

• Performed detailed analyses on the impact of various parameters such as sensor noise, and navigation drift, providing valuable insights into optimizing system performance.

• Conducted extensive experiments and simulations on multiple underwater environments to validate the effectiveness of the proposed approach.

**SurgiQ Srl** — Genoa, Italy
*Bachelor Thesis: AI-based scheduling in healthcare* — *Mar 2020 - Jul 2020*

• Developed and implemented an intelligent scheduling system for a hospital, resulting in 97% efficiency.

• Conducted thorough research on inductive AI techniques to inform the development of the scheduling system, leading to its publication on AI*IA and CLIC.

## SKILLS

**Programming**: Python, Docker, Google Cloud Platform (GCP), Git
**Deep Learning Frameworks:** PyTorch, JAX, TensorFlow, Keras
**Data Manipulation and Analysis:** Pandas, MongoDB, SQL
**Languages**: English, Italian

## PROJECTS

**Real-time object detection | KTH** — Stockholm, Sweden
*Core skills: Python* — *May 2022 - Jul 2022*

• Managed intensive experiments by training deep neural networks utilizing Python, YOLOv5, reaching 45 FPS and 61% mAP for real-time object detection.

• Optimized DNN model to identify 4 different classes on 10K labeled image

• https://github.com/marcoschouten/yolov5-real-time-object-detection

**Imitation learning for skill transfer | TU Delft** Delft, Netherlands

*Core skills: Python* *Sep 2021 - Dec 2021*

• Developed a probabilistic model that transfers hand movements to manipulator motions by imitation learning.

• Implemented a Gaussian Mixture Model (GMM) to characterize task trajectory with over 90% accuracy, resulting in an optimal internal representation of geodesic movement.

• https://github.com/marcoschouten/imitation-learning

**Training time estimation for CNN | TU Delft** Delft, Netherlands

*Core skills: Python, Docker, GCP* *Sep 2021 - Nov 2021*

• Utilized predictive modeling, linear regression and ANOVA to optimize CNN training jobs, completing 33% more within a pre-defined timeline utilizing 4 nodes and 16 GB of RAM

• https://github.com/marcoschouten/quantitative-performance-evaluation-for-computing-systems

**Starcraft AI Bot Competition | KTH** Stockholm, Sweden

*Core skills: C++, C#, Unity3D* *Apr 2021 - Jun 2021*

• Developed behaviour tree algorithm in C++ & BWAPI for controlling multi-agent Starcraft AI, resulting in a 50% win rate with 50+ simulation matches.

• https://github.com/marcoschouten/multi-agent-systems

## AWARDS

• Selected at Nova, a top 3% merit-based talent network for career development

• Hattrick Award in Artificial Intelligence at KTH, awarded to 20% of the students

• Selected at Lead the Future, mentorship with a 20% acceptance rate nationwide

• Scholarship at University of Genoa 2018, 2019,2020, ranked 4th out of 150+ students

## PUBLICATIONS

(1) Cademartori, L., Galatà, G., Monaco, C.L., Maratea, M., Mochi, M. and **Schouten, M.**, 2022. An ASP-based Approach to Master Surgical Scheduling. CILC 2022: 37th Italian Conference on Computational Logic.

(2) Alviano, M., Bertolucci, R., Cardellini, M., Dodaro, C., Galatà, G., Khan, M., Maratea, M., Mochi, M., Morozan, V., Porro,I. and **Schouten, M**. 2020. Answer Set Programming in Healthcare: Extended Overview. IPS-RCRA@ AI*IA